

Advanced Hydrology Final Project Guidelines

1. Project Objective

The final project is designed to help students develop a complete data-analysis workflow for a realistic hydrologic, meteorological, or environmental problem using the concepts introduced in this course.

Rather than mechanically applying predefined methods, students are expected to:

- identify a meaningful applied problem,
- investigate the characteristics of the data,
- justify methodological choices,
- evaluate model performance and uncertainty,
- and critically interpret the results from hydrologic or engineering perspectives.

Projects should emphasize:

- problem-driven analysis,
- critical thinking,
- interpretation of uncertainty and limitations,
- and appropriate integration of statistical, time series, spatial, or data assimilation methods.

Projects are conducted in teams of 3 students.

2. General Requirements

Regardless of the selected methods, projects should include:

Clear problem formulation

Students should clearly explain:

- What is the target problem?
- Why is it important?
- What are the practical or scientific motivations?

Examples:

- Predicting short-term wind speed for hazard warning
- Estimating extreme rainfall return levels
- Understanding spatial rainfall variability
- Forecasting streamflow or water levels

Exploratory analysis of data characteristics

Students should investigate the characteristics of the data before selecting methods.

Examples:

- Temporal or spatial patterns
- Trends or seasonality
- Correlation structure
- Distributional characteristics
- Missing data or noise issues
- Extreme events or outliers
- Changes in variability over time

(Note: Not every project is expected to include all of the above.)

Justification of selected methods

Students should explain:

- Why the selected methods are appropriate
- What assumptions are involved
- Why alternative approaches may or may not be suitable

Examples:

- Using transformations to reduce skewness
- Selecting ARMA models for autocorrelated data
- Using PCA to reduce dimensionality
- Applying Kriging because of spatial dependence
- Using EnKF to improve forecast updating

Model calibration and validation

Projects should include appropriate calibration and validation procedures.

Examples:

- Parameter estimation
- Residual analysis
- Forecast evaluation
- Cross-validation

- Comparison between observed and simulated data
- Evaluation using RMSE, correlation, likelihood, or other metrics

Uncertainty analysis and discussion

Students should discuss uncertainty and limitations of their analysis.

Examples:

- Sensitivity to parameter choices
- Distribution-selection uncertainty
- Forecast uncertainty
- Spatial interpolation uncertainty
- Effects of limited data length
- Model structural limitations

Interpretation of hydrologic or engineering implications

Students should interpret results beyond statistical calculations.

Examples:

- Implications for hazard assessment
- Forecast reliability
- Infrastructure design considerations
- Environmental interpretation
- Physical meaning of identified patterns

Students are encouraged to select methods based on the characteristics of their data and objectives.

Possible methods may include:

- probability distributions and extreme value analysis,
- regression models,
- transformed variables,
- time series analysis,
- multivariate analysis,
- PCA,

- spatial interpolation and geostatistics,
- spatio-temporal modeling,
- Kalman filtering or data assimilation,
- or other relevant approaches.

3. Suggested Project Topics

Students may choose one of the following example topics, modify them, or propose their own topic subject to instructor approval.

A. Extreme Value Analysis and Reliability Estimation

Examples:

- Maximum wind speed
- Extreme rainfall
- Flood peak analysis

Possible analyses:

- Distribution fitting (Gumbel, GEV, Lognormal, etc.)
- Return-period estimation
- Reliability analysis
- Threshold exceedance probability
- Sensitivity to distribution choice
- Uncertainty evaluation

B. Forecasting, Time Series Modeling, and Simulation

Examples:

- Wind speed forecasting
- Rainfall prediction
- Temperature or pressure modeling
- Streamflow or groundwater-level analysis

Possible analyses:

- Regression models
- Variable transformation

- AR / MA / ARMA models
- Residual analysis
- Forecast evaluation
- Time series simulation
- State-space modeling
- Kalman Filter or Ensemble Kalman Filter (optional)
- Comparison of predictive methods

C. Multivariate Analysis of Environmental Variables

Examples:

- Wind–rainfall relationships
- Rainfall–runoff relationships
- Multi-station atmospheric analysis

Possible analyses:

- Correlation analysis
- Multiple regression
- VAR models
- PCA
- Dependency analysis

D. Spatial or Geostatistical Analysis

Examples:

- Spatial rainfall interpolation
- Wind-field simulation
- Groundwater spatial variability

Possible analyses:

- Variograms
- Kriging
- Spatial simulation
- Spatial uncertainty quantification

E. Spatio-Temporal Analysis and Forecasting

Examples:

- Regional precipitation forecasting
- Pressure-field evolution
- Weather-event evolution

Possible analyses:

- Spatio-temporal regression
- PCA-based temporal analysis
- Coupled spatial-temporal models
- Uncertainty propagation

F. Student-Proposed Topics

Students are encouraged to propose their own topics if they are relevant to hydrology, meteorology, environmental systems, or related engineering applications.

4. Data

Students may use any open-source dataset appropriate for their project.

Possible sources include:

- Central Weather Administration (Taiwan)
- River or reservoir monitoring data
- Reanalysis datasets
- International meteorological or hydrologic databases

Data sources are not limited to Taiwan.

Students are responsible for:

- explaining the dataset,
- checking data quality,
- and appropriately preprocessing the data.

5. Suggested Difficulty Adjustment

Projects involving more advanced methodologies may receive additional basic scores depending on methodological complexity.

Suggested difficulty adjustment:

- +3: spatial or geostatistical analysis
- +5: advanced spatio-temporal, multivariate, or data assimilation methods

The adjustment depends on:

- methodological difficulty,
- technical depth,
- and quality of implementation and interpretation.

6. Project Deliverables

- Slides for oral presentation (no written report required)
- Presentation time:
 - ~12 minutes presentation
 - ~1–3 minutes Q&A

Requirements:

- All team members must actively participate.
- All team members should speak during the presentation.
- Slides should contain clear visualizations and figures.

7. Evaluation Rubric

Criterion	Weight	Key Aspects
Content Understanding	30%	Understanding of hydrologic/statistical concepts and appropriate methodological application
Organization & Structure	15%	Logical flow, clarity, and time management
Depth of Analysis	25%	Critical thinking, uncertainty discussion, interpretation, and methodological justification
Visual Aids	15%	Quality and clarity of figures and slides
Delivery & Engagement	10%	Speaking clarity and audience engagement
Response to Questions	5%	Ability to answer questions thoughtfully

Total = 100%

8. Grading Structure

Rehearsal Presentation: 40%

- Instructor evaluation: 60%
- Peer evaluation from classmates: 30%
- Peer review completion: 10%

Final Presentation: 60%

- Instructor and TA evaluation only

Additionally:

- In-group teamwork peer review will be used to adjust individual scores based on team contribution.

9. Suggested Presentation Structure

- Title and Motivation
- Problem Definition and Objectives
- Data and Exploratory Analysis
- Methods and Model Selection
- Calibration and Validation
- Results and Visualizations
- Uncertainty and Limitations
- Hydrologic/Engineering Interpretation
- Conclusions

10. Timeline

Task	Deadline
Final Project Guideline Released	May 7
Preliminary Topic Selection	May 14
Data Collection and Proposed Analysis Methods	May 21
Rehearsal Presentation (~80% completed)	June 4
Final Presentation & Slide Submission	June 11

11. Important Notes

- Active participation is mandatory.
- Students must complete peer evaluations and teamwork evaluations.
- Academic honesty is required.
- Proper citation of external data and sources is required.
- Topic selection may be adjusted later with instructor approval.